Docon

Prepared by

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Abstract

Docon is an digital healthcare platform designed to simplify patient management, telemedicine consultations, and medical record-keeping for doctors, and hospitals. This system integrates online appointment scheduling, electronic health records (EHR), e-prescriptions, and remote consultations into a seamless and secure interface. By leveraging advanced technology, Docon enhances efficiency in healthcare delivery, improves patient engagement, and streamlines doctor-patient interactions.

This project aims to bridge the gap between patients and healthcare providers, enabling accessible, efficient, and data-driven medical care.

Objectives of the Project

The primary objective of the Docon digital healthcare platform is to streamline and enhance the delivery of healthcare services by integrating advanced digital tools for patient management, telemedicine, electronic health records (EHR), and e-prescriptions. The project aims to bridge the gap between patients and healthcare providers by enabling a secure, accessible, and data-driven environment that fosters efficient medical care, improves patient engagement, and supports informed clinical decision-making.

Project Goals

- Enhanced healthcare accessibility through telemedicine.
- Efficient patient management with secure digital health records.
- Reduced waiting times and improved doctor-patient interactions.

Core Functionalities

- Online Appointment Scheduling & Queue Management

Appointment booking system with time slot optimization.

Real-time updates and notifications for upcoming consultations.

- Electronic Health Records (EHR) & e-Prescriptions

Secure storage of patient medical history and prescriptions.

Insights for diagnosis and treatment recommendations.

Seamless sharing of medical data with authorized doctors.

- Health Insights & Reminders

Personalized health recommendations based on medical history.

Medication reminders and follow-up alerts.

Predictive analytics for chronic disease management.

- Billing

Automated payment processing.

Integration with third-party payment gateways for seamless transactions.

- Telemedicine & Remote Consultations

Chat-based virtual doctor consultations.

Symptom checker for Preliminary assessments.

scope

- Doctors

Manage patient records and medical history digitally.

Conduct remote consultations via chat.

Generate and share digital prescriptions securely to patient.

- Patients

Book online or in-clinic doctor appointments.

Access personal medical records and prescriptions.

- Clinics & Hospitals

Automate patient data management and billing.

Optimize appointment scheduling and queue management.

Technologies

- Technology:- MERN
- Fronted:-React js, vite
- Backend:-Node js , Express js
- Database:-MongoDB
- Server:-Express server

Workflow

Patients register and book consultations through the platform.

Doctors review patient history and conduct in-person or virtual consultations.

Digital prescriptions and medical records are securely stored and shared.

Patients receive follow-up reminders and health insights for better care management.

Implementation

System Architecture & Design

- Backend: Node.js | Frontend: React
- Responsive UI for desktop

User & Data Management

- Role-based access for doctors, admin, patients
- JWT for secure login

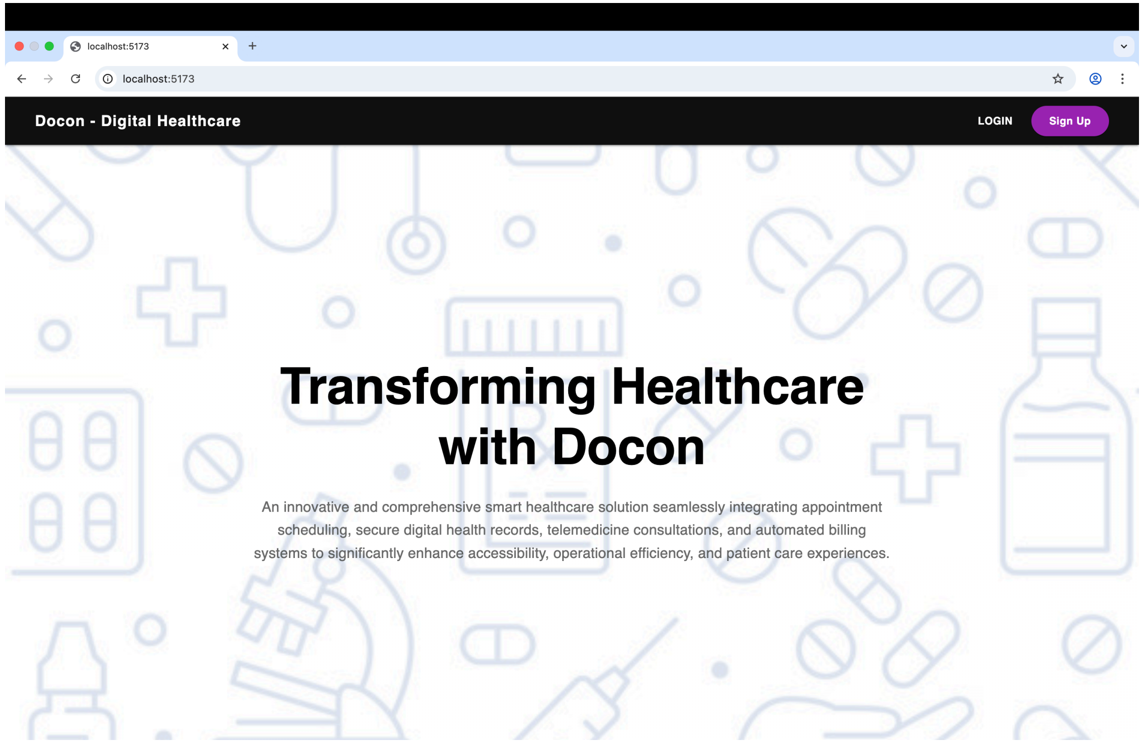
Key Features

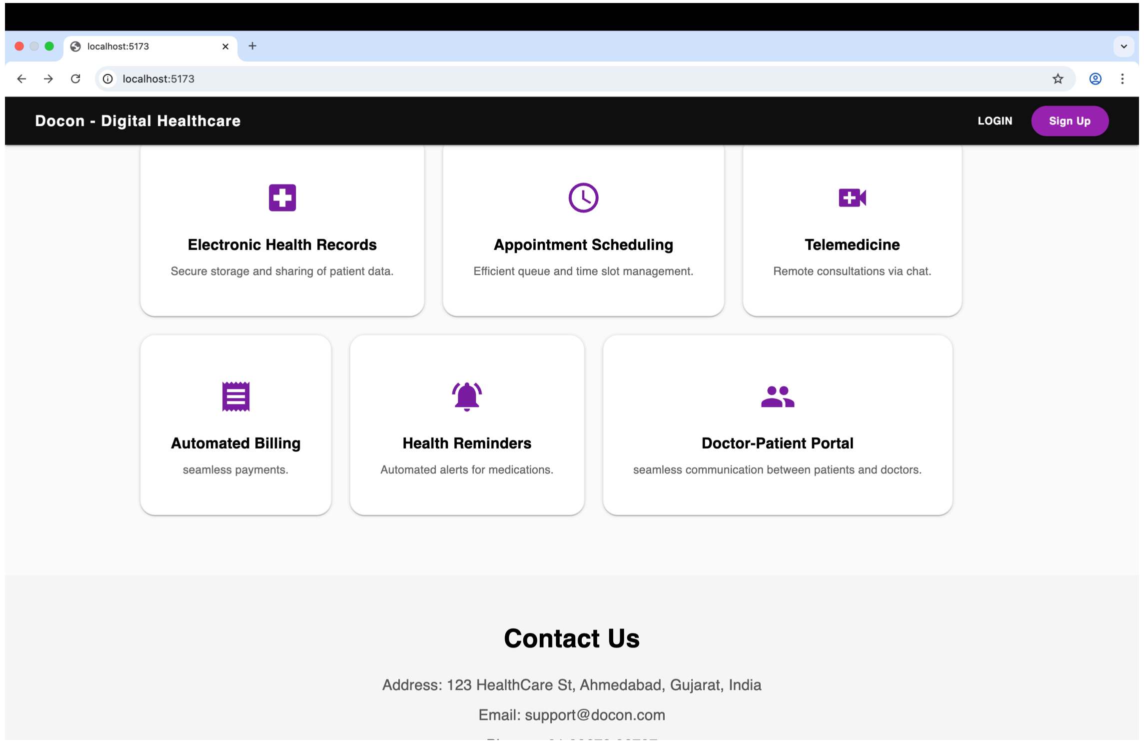
- Online appointment scheduling with real-time
- Telemedicine via secure chat
- Digital prescriptions
- EHR for centralized medical records

Interoperability & Analytics

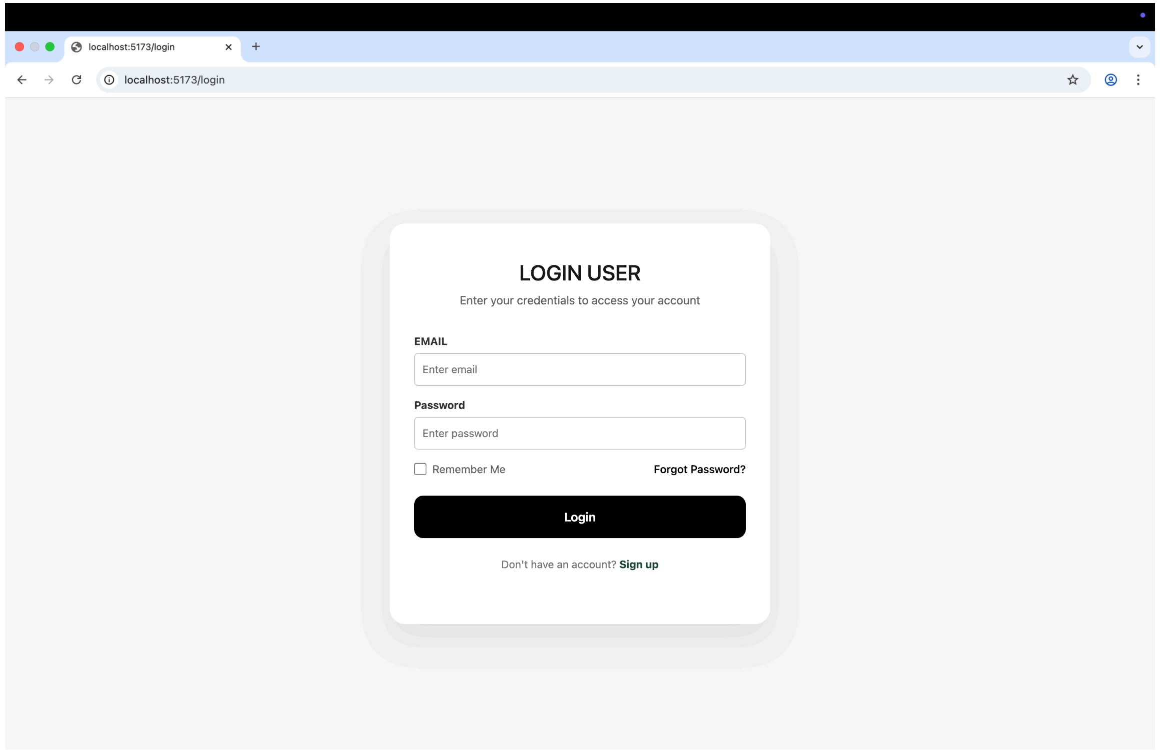
- Dashboard for doctors, admin, patients

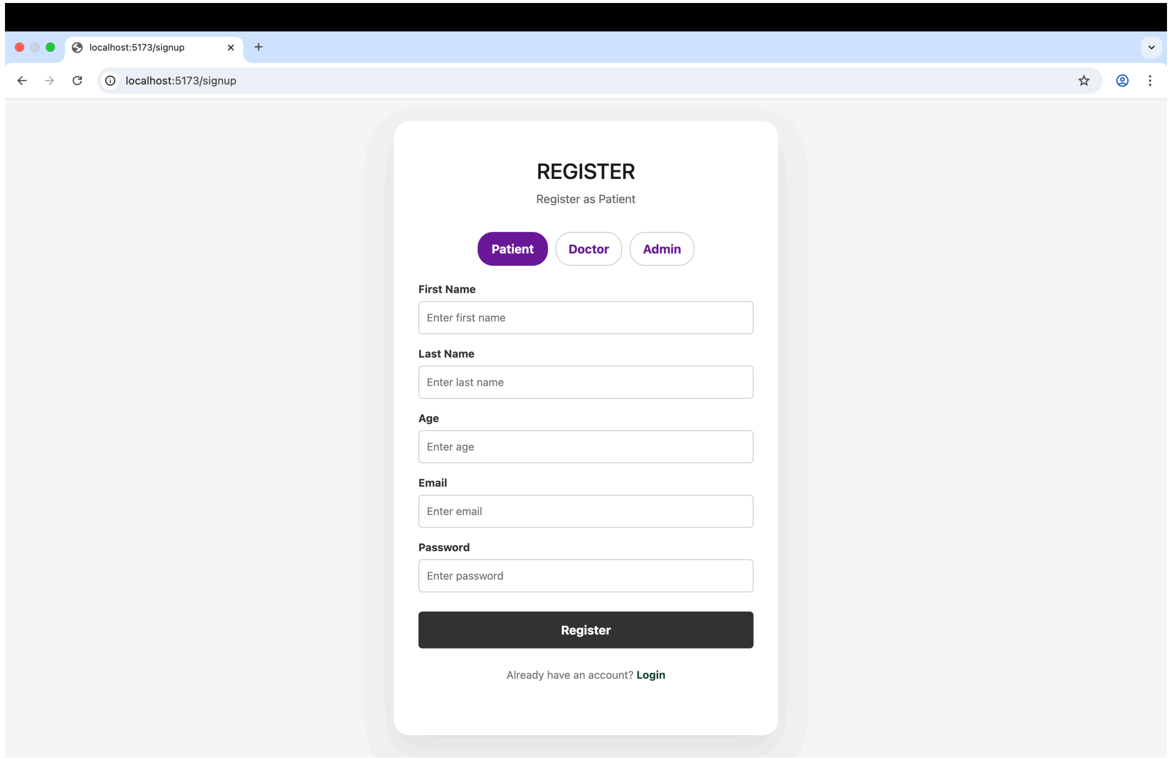
Result screen shots





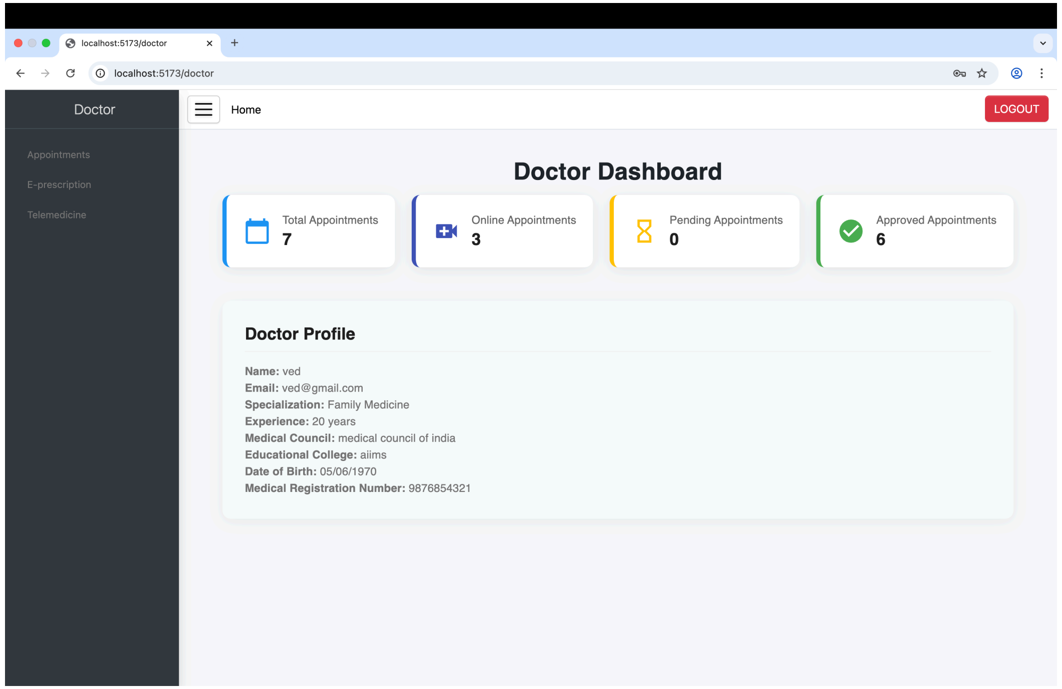
Login & register for doctor, patient, admin

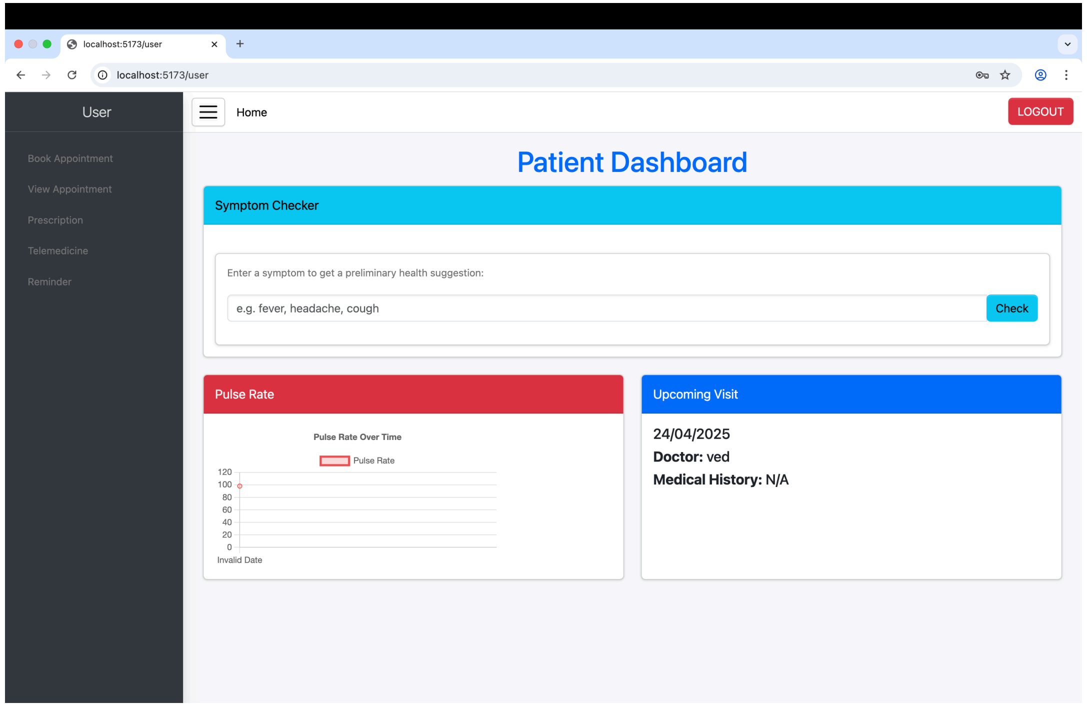




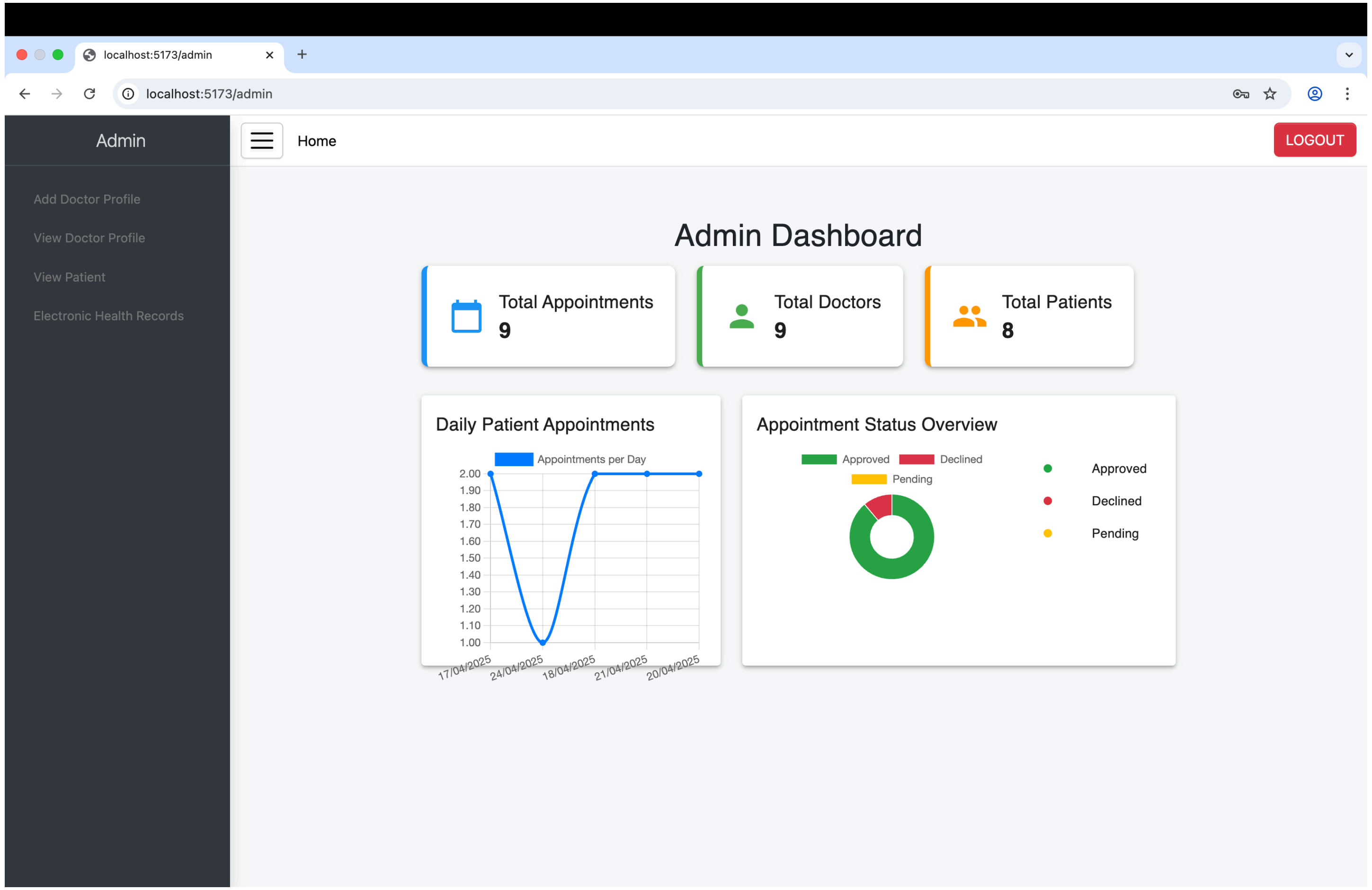
Allows users (doctors, admin and patients) to securely access the platform by logging in or creating a new account with essential credentials.

Dashboard



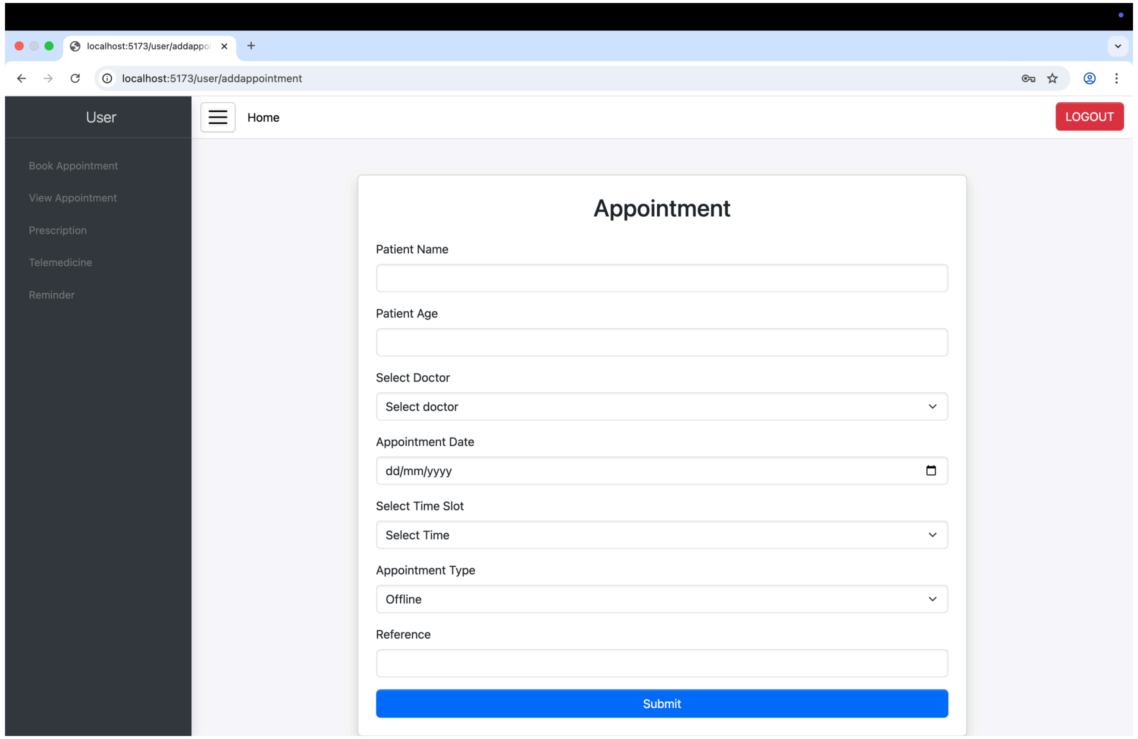


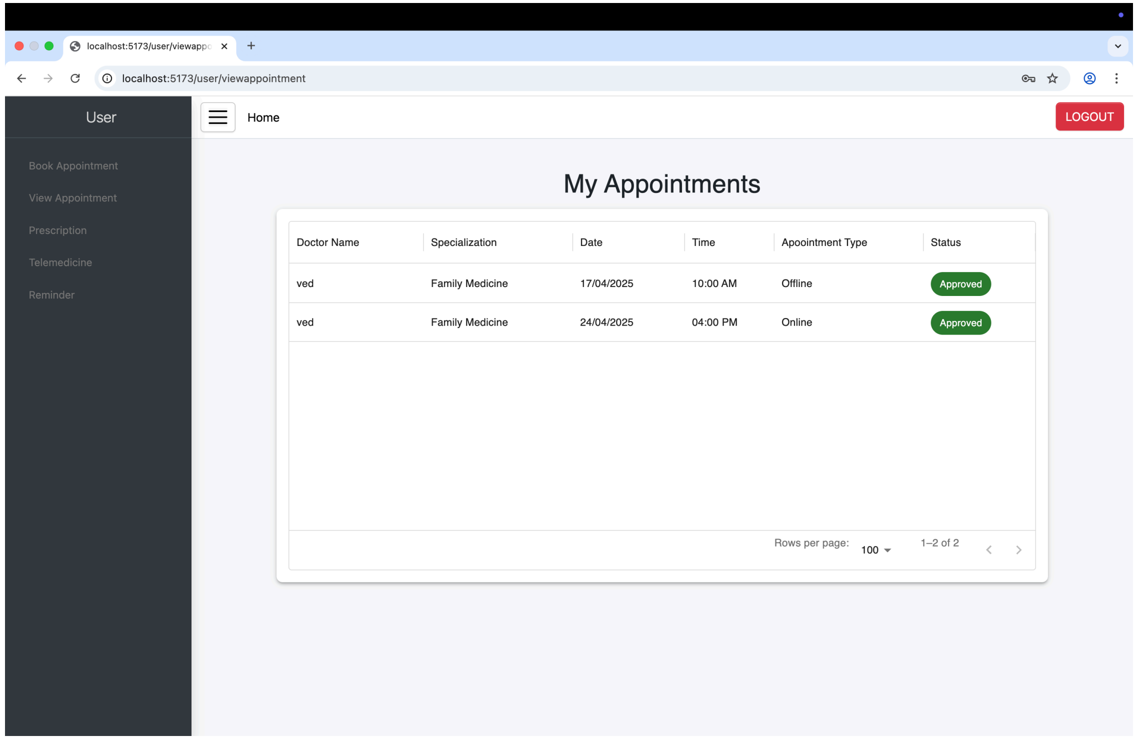
Provides doctors with tools for appointments details , and doctor profile, while patients can view symptom checker, pulse rate.



The admin dashboard displays key metrics: Total Appointments, Total Doctors, Total Patients, and Daily Patient Appointments for quick insights. An Appointment Status Overview Chart visualizes statuses like *Pending*, *approved*, and *Cancelled* to monitor performance at a glance.

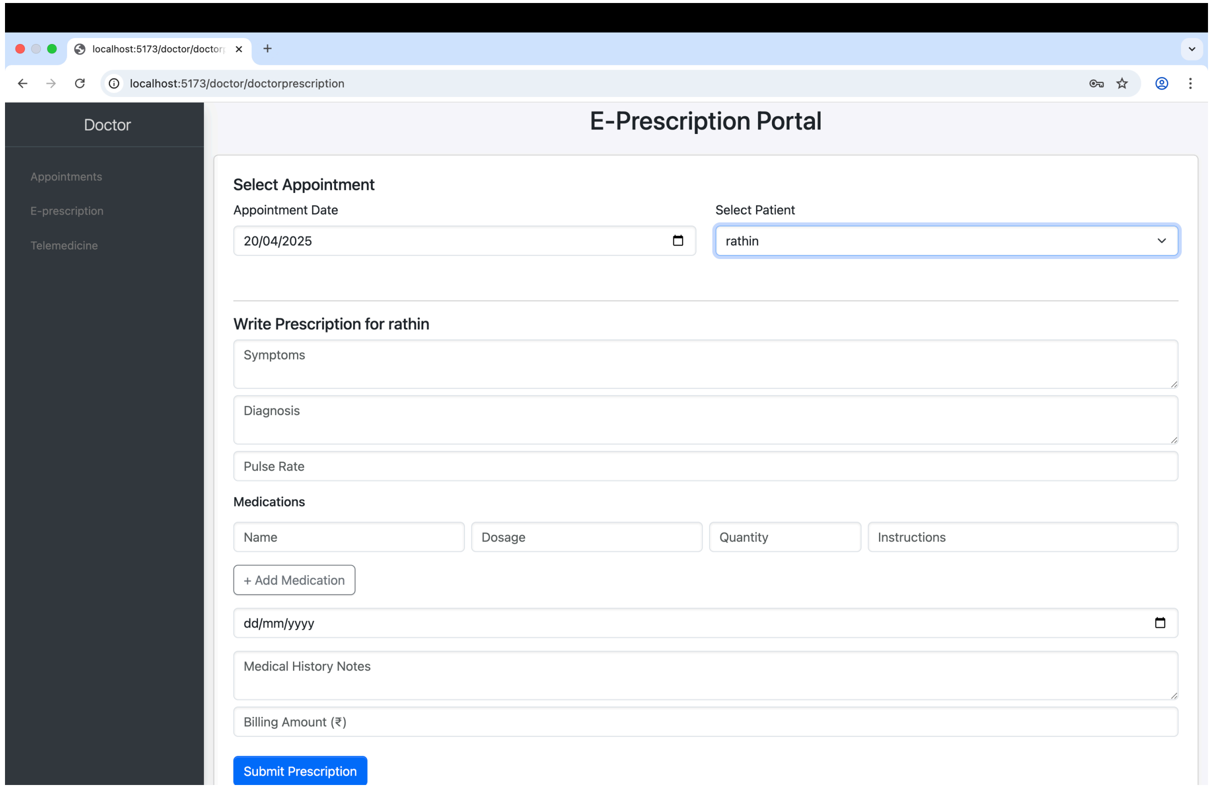
Booking appointment

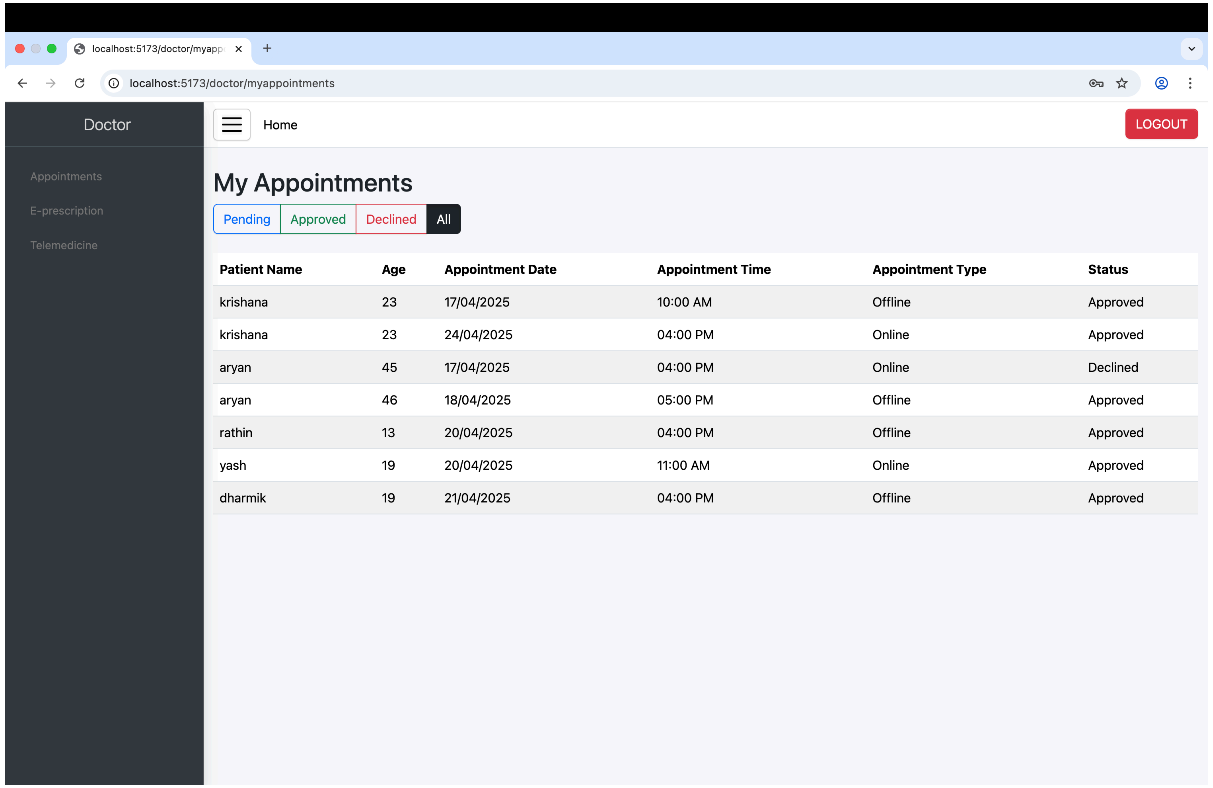




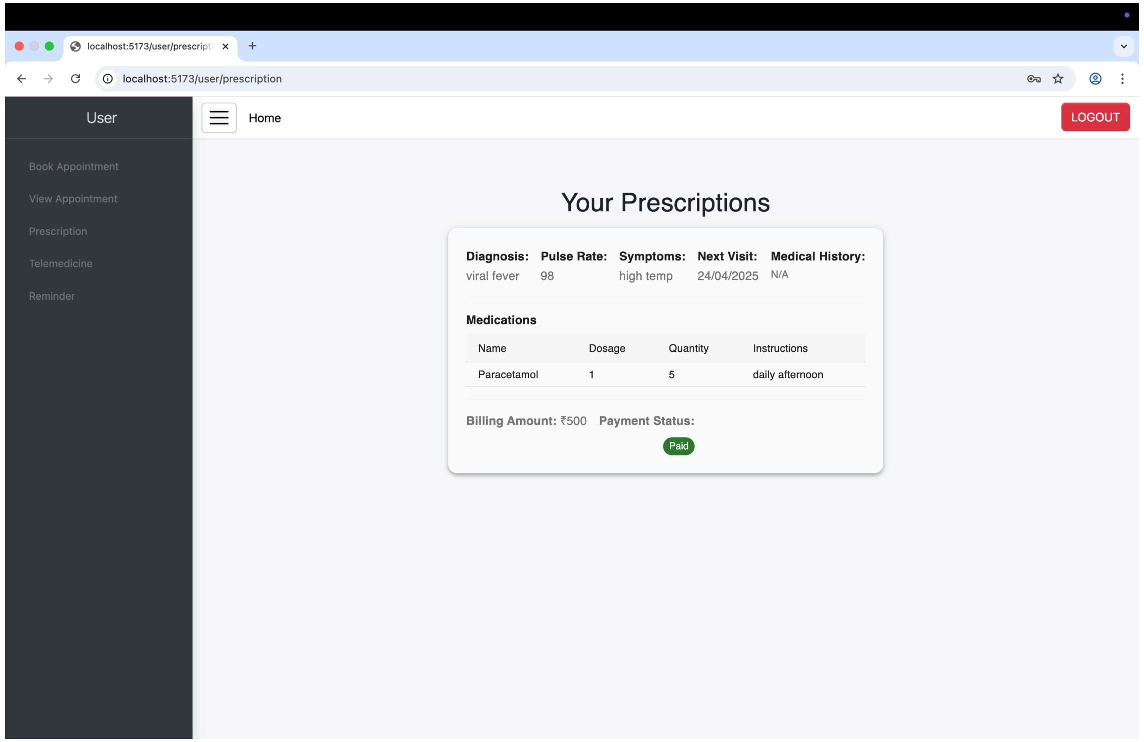
The Book Appointment page allows users to select a doctor, date, and time to schedule a medical visit. The View Appointment Status page shows upcoming and past appointments with real-time status updates like Pending, approved,or declined.

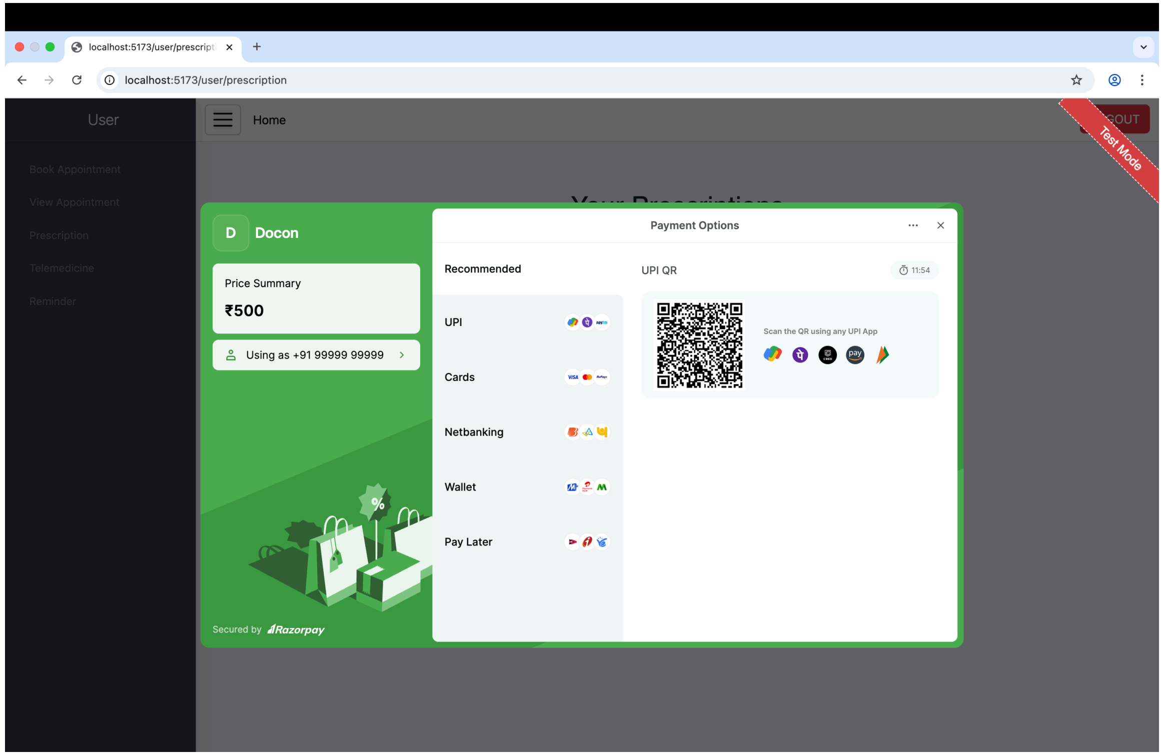
Doctor appointments and e-prescription



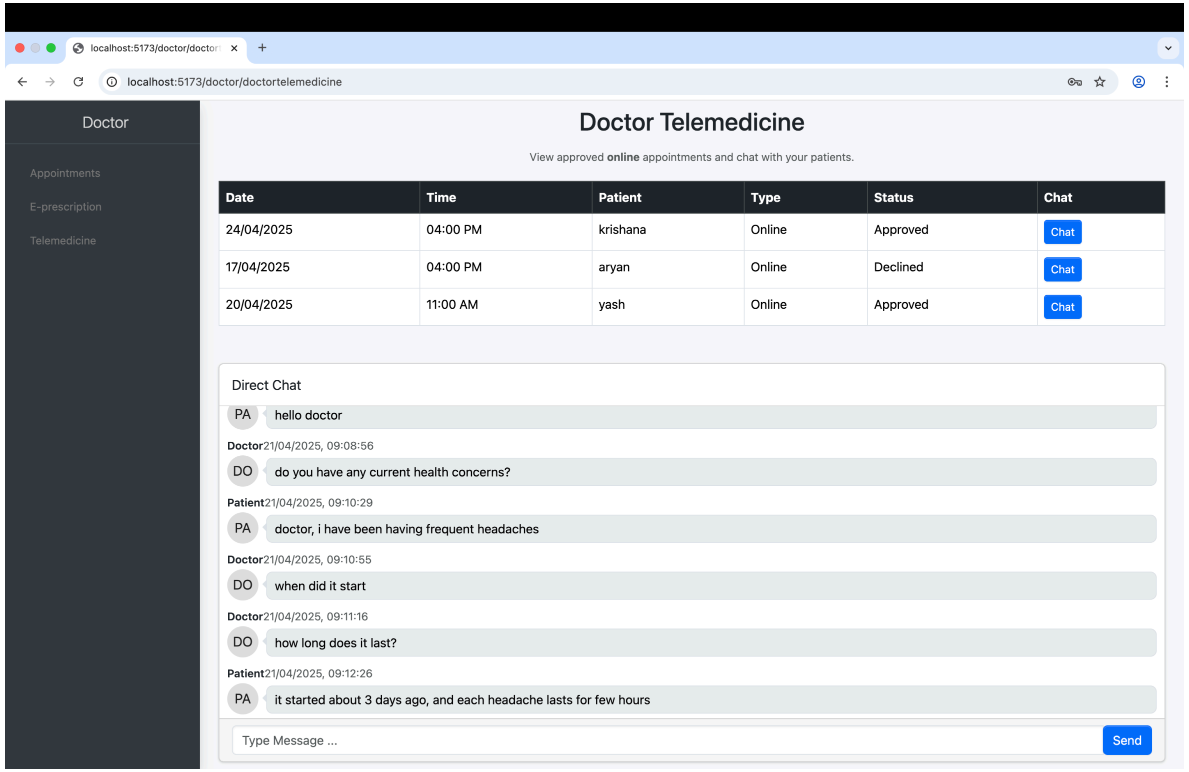


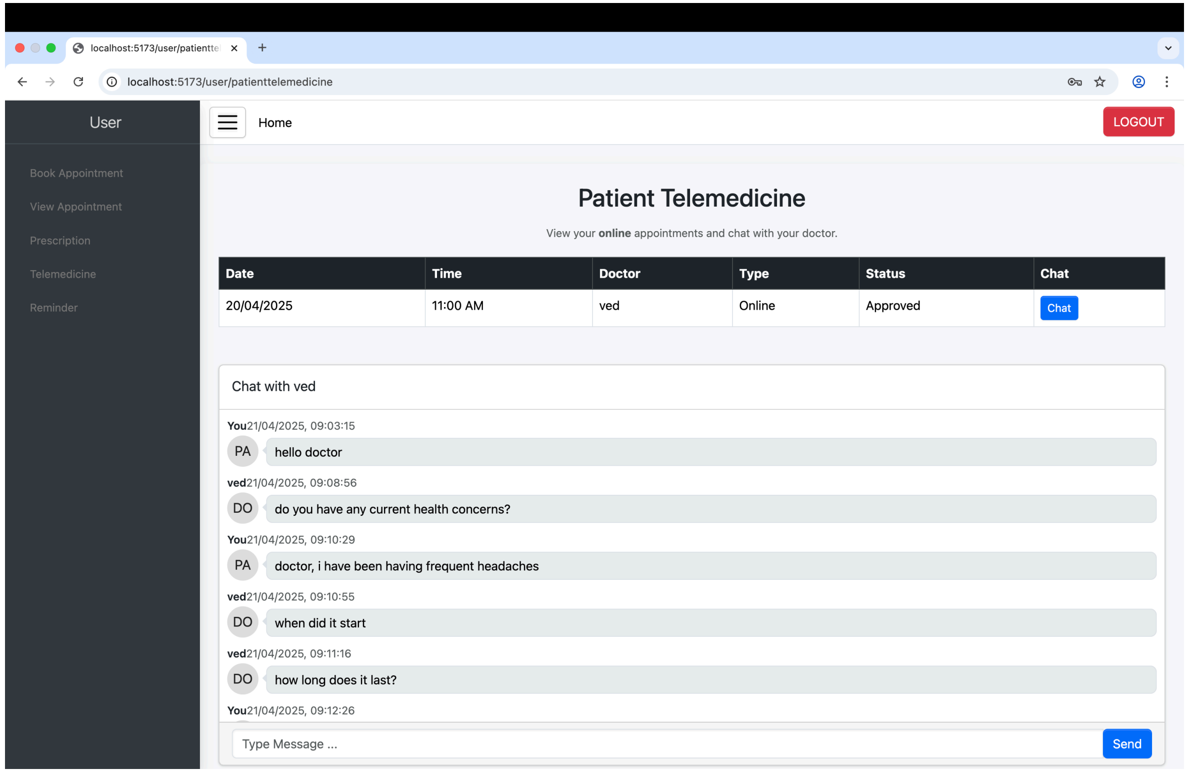
User prescription and payment gateway



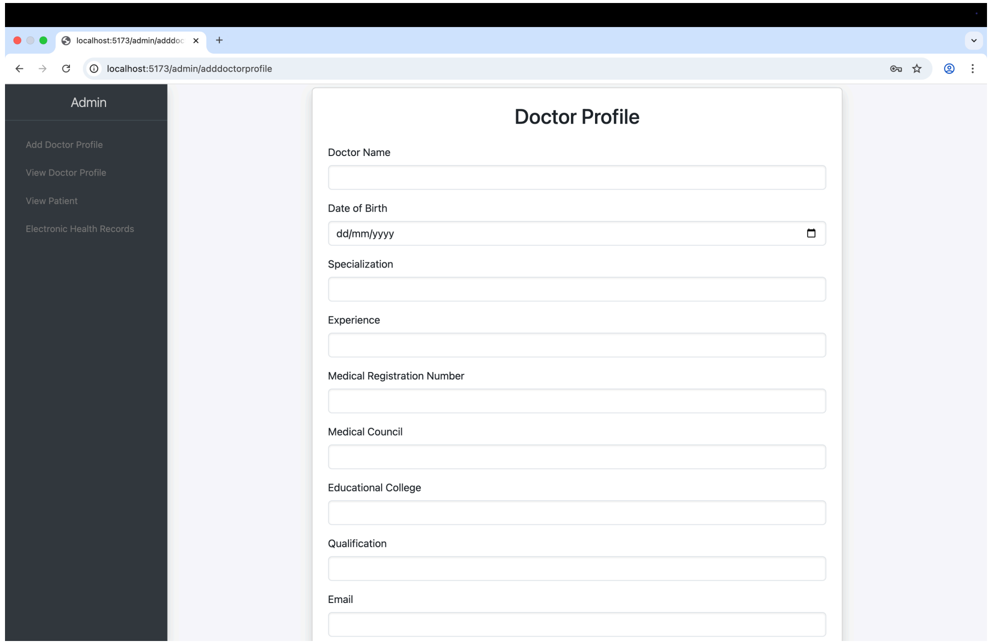


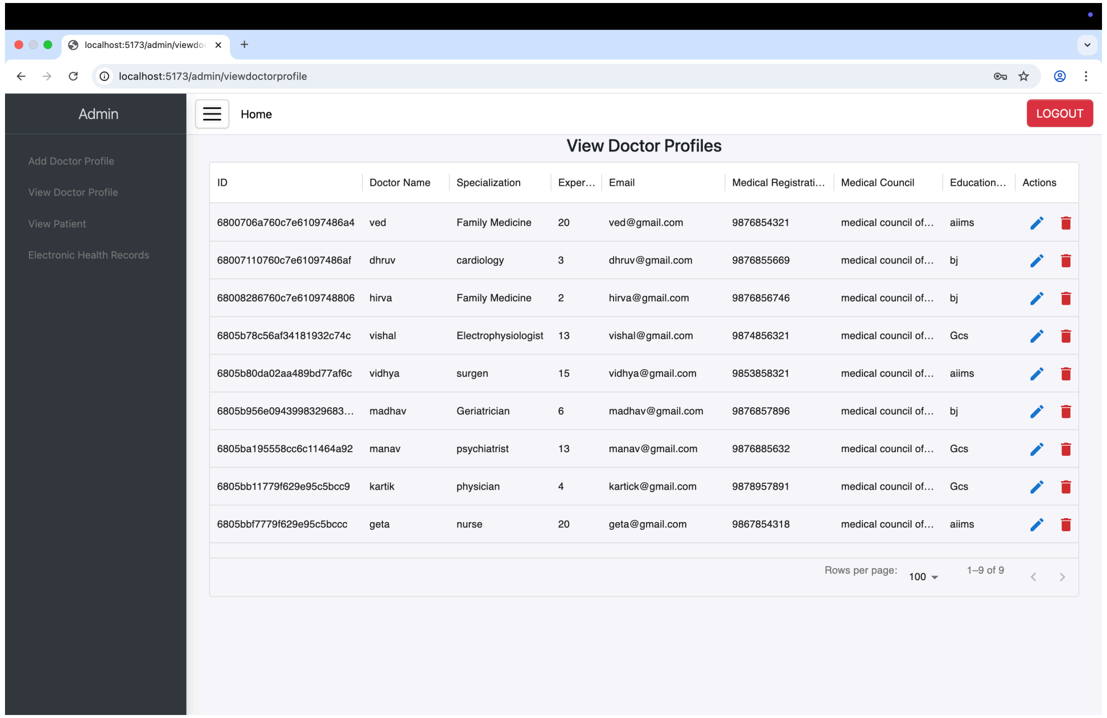
Chat base Telemedicine



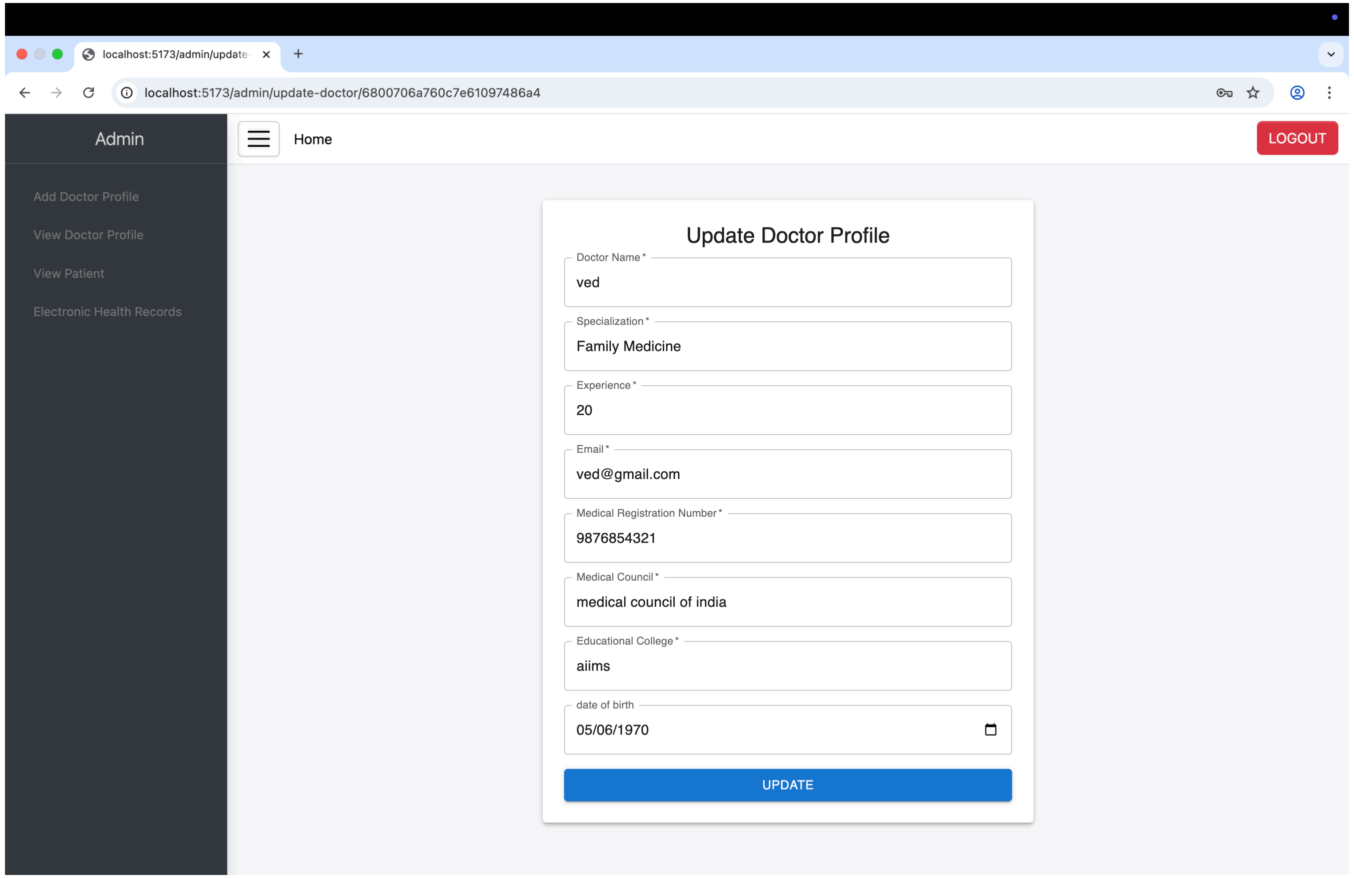


Doctor profile add and view

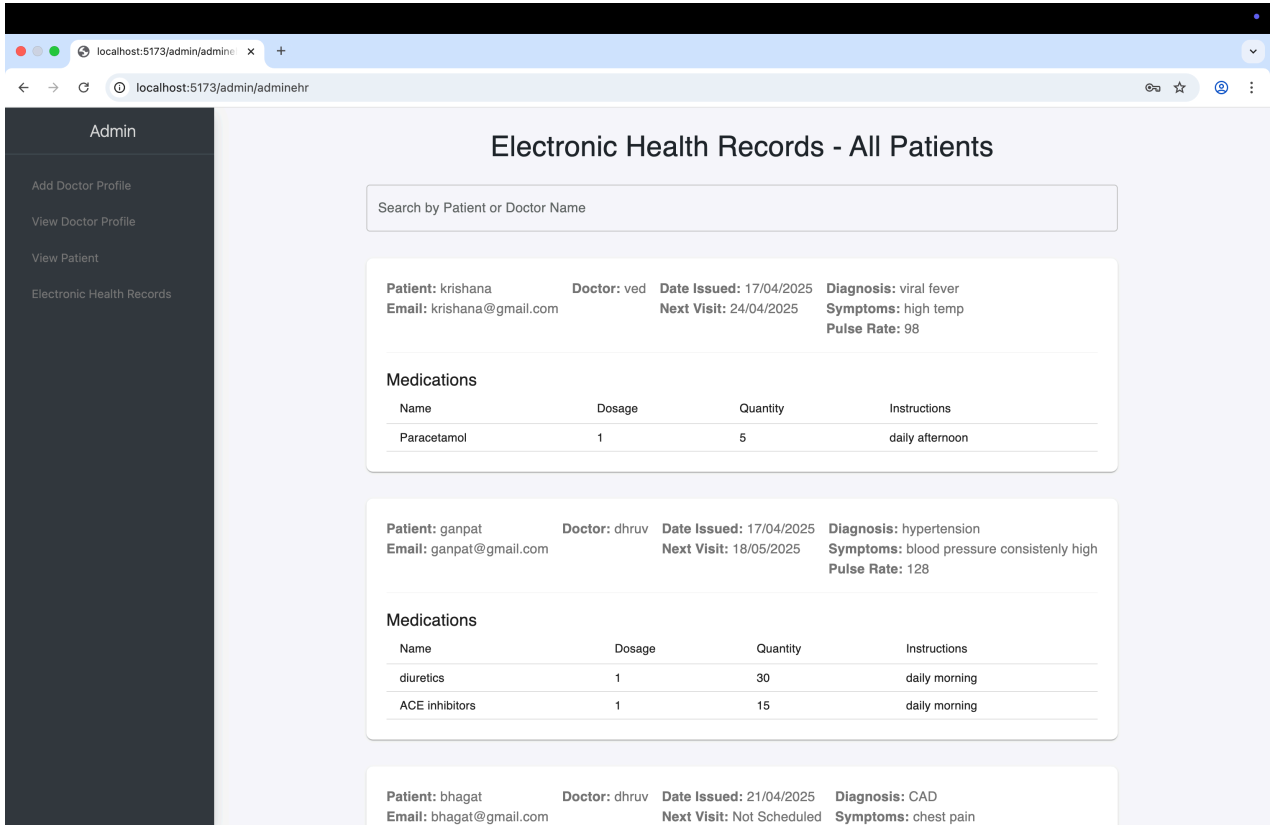




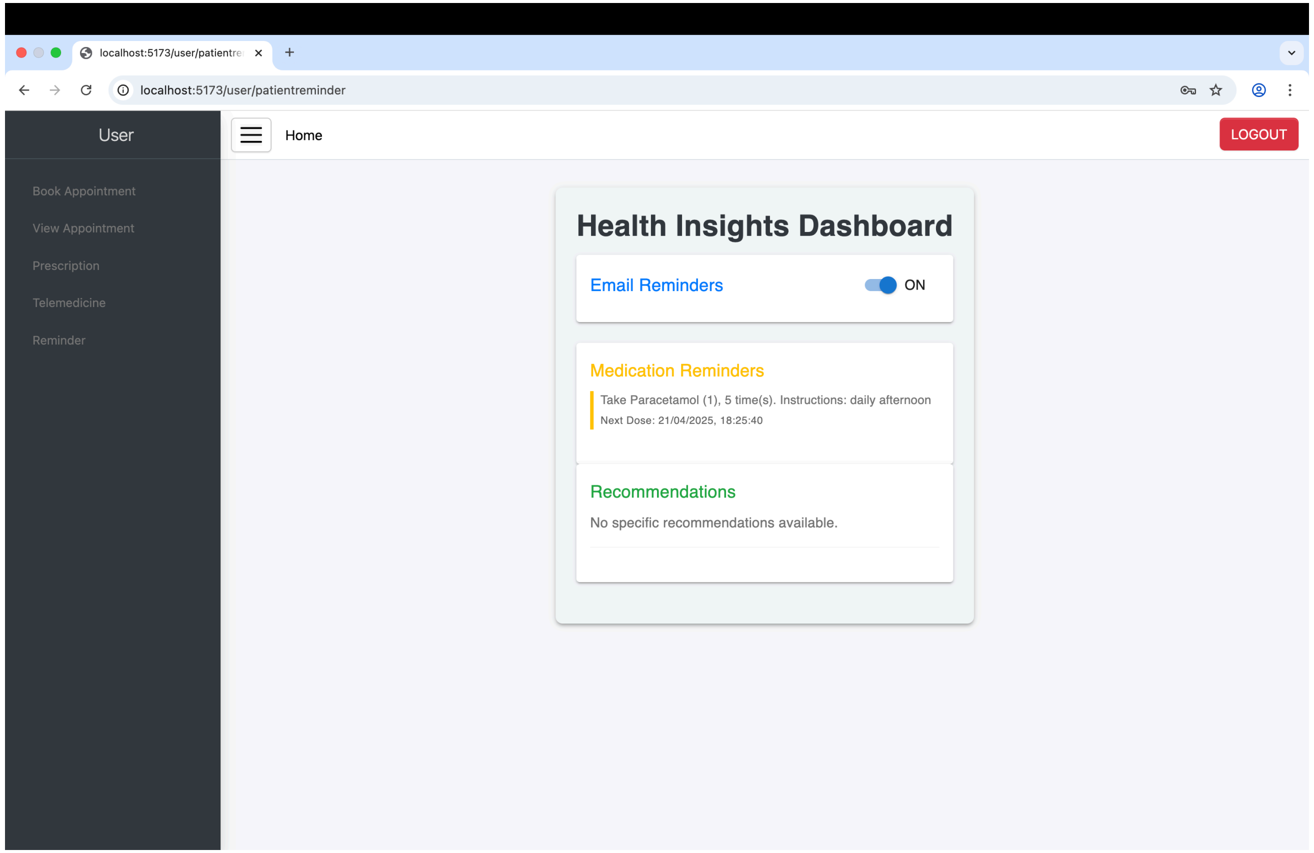
Update doctor Profile



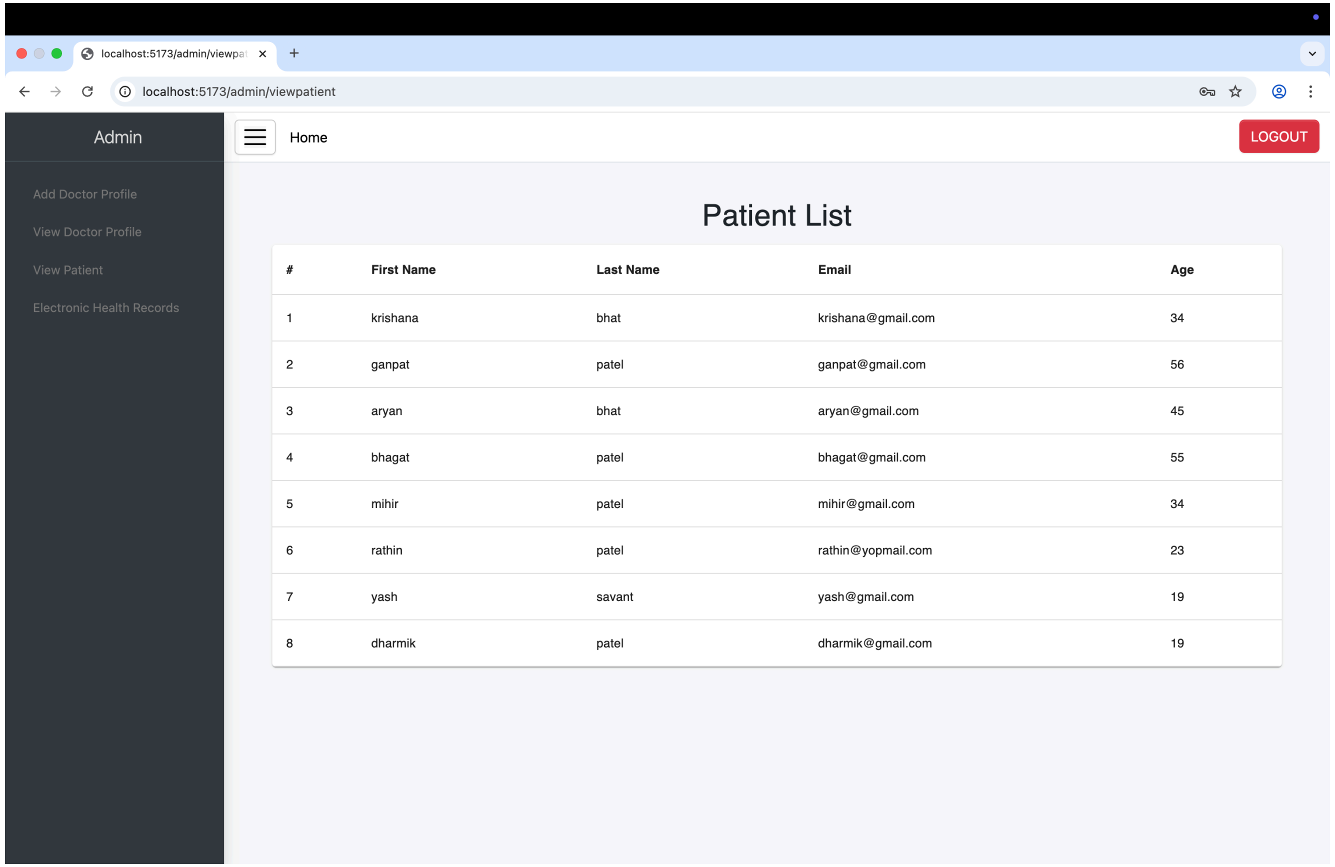
Electronic health records



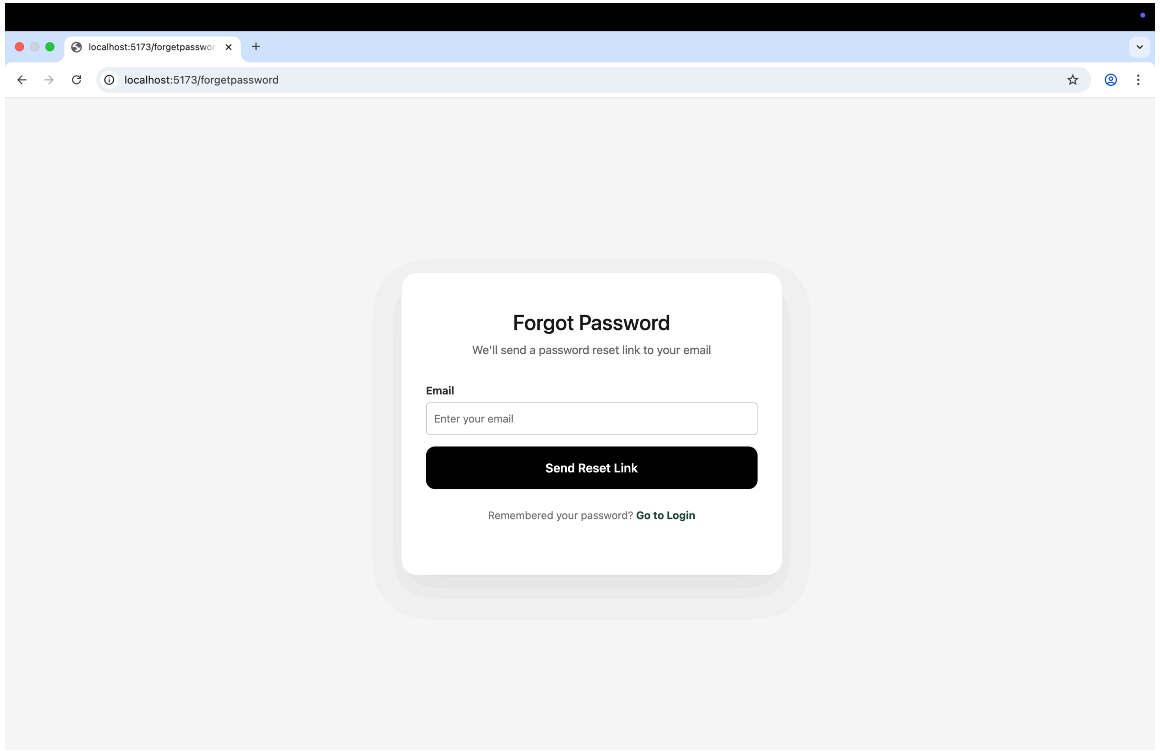
Health reminder

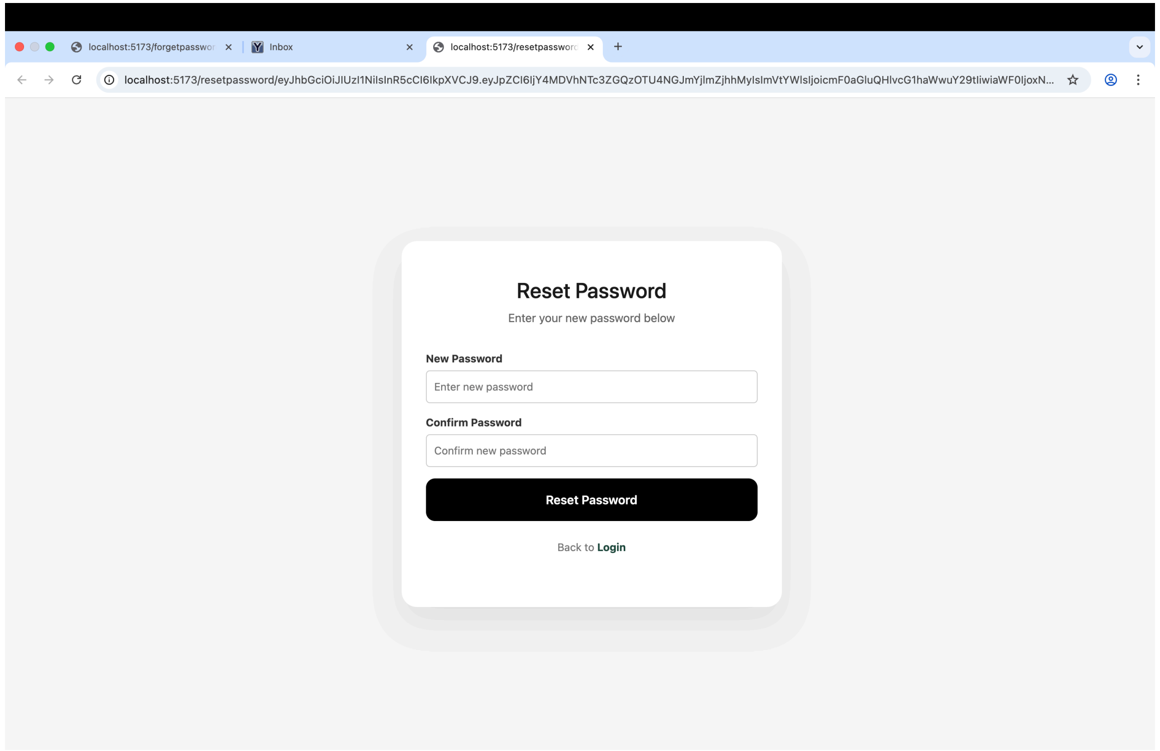


Register Patient list



Reset password





The Forgot Password page lets users request a password reset link by entering their registered email. The Reset Password page allows users to securely set a new password using the link sent to their email.

Conclusion

Docon simplifies and modernizes healthcare delivery

By integrating appointments, EHR, telemedicine, and e-prescriptions into one secure platform, Docon enhances efficiency for doctors and accessibility for patients.

A step toward smarter, data-driven healthcare

digital workflows, and seamless integrations, Docon empowers healthcare providers to deliver better, faster, and more personalized care.

Scalable, secure, and future-ready

Built on robust cloud architecture with a focus on privacy and interoperability, Docon is designed to grow with evolving healthcare needs.

Future Enhancement of Implementation

- Enhanced healthcare accessibility through telemedicine.
- Efficient patient management with secure digital health records.
- Reduced waiting times and improved doctor-patient interactions.
- Open APIs for lab, pharmacy, insurance, and IoT health devices.
- AI-driven suggestions for follow-ups & treatment insights.

References

- **MongoDB**:

MongoDB, Inc. "MongoDB Documentation." Available at: https://docs.mongodb.com/ MongoDB is the NoSQL database used for storing doctor profile data, user information, and patient history in the project.

- **Express.js**:

Express.js Team. "Express Documentation." Available at: https://expressjs.com/ Express is used for creating the backend API, handling HTTP requests, routes, and middleware for the hospital management system.

- **Node.js**:

Node.js Foundation. "Node.js Documentation." Available at: https://nodejs.org/en/docs/ Node.js is the runtime environment used for the backend server to execute JavaScript code.

- **React.js**:
  - React Team. "React Documentation." Available at: https://reactjs.org/docs/getting-started.html React is used for building the user interface of the Docon.